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Exploring genomic feature selection: A comparative analysis of GWAS and machine learning algorithms in a large-scale soybean dataset

Hawlader A. Al-Mamun¹  | Monica F. Danilevicz¹ | Jacob I. Marsh² | Cedric Gondro³ | David Edwards¹ 

¹Centre for Applied Bioinformatics, and School of Biological Sciences, University of Western Australia, Perth, Western Australia, Australia

²Department of Biology, University of North Carolina, Chapel Hill, North Carolina, USA

³Department of Animal Science, Michigan State University, East Lansing, Michigan, USA

Correspondence

David Edwards, Centre for Applied Bioinformatics, and School of Biological Sciences, University of Western Australia, Perth, WA 6009, Australia.
Email: dave.edwards@uwa.edu.au

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Abstract

The surge in high-throughput technologies has empowered the acquisition of vast genomic datasets, prompting the search for genetic markers and biomarkers relevant to complex traits. However, grappling with the inherent complexities of high dimensionality and sparsity within these datasets poses formidable hurdles. The immense number of features and their potential redundancy demand efficient strategies for extracting pertinent information and identifying significant markers. Feature selection is important in large genomic data as it helps in enhancing interpretability and computational efficiency. This study focuses on addressing these challenges through a comprehensive investigation into genomic feature selection methodologies, employing a rich soybean (*Glycine max* L. Merr.) dataset comprising 966 lines with over 5.5 million single nucleotide polymorphisms. Emphasizing the “*small n large p*” dilemma prevalent in contemporary genomic studies, we compared the efficacy of traditional genome-wide association studies (GWAS) with two prominent machine learning tools, random forest and extreme gradient boosting, in pinpointing predictive features. Utilizing the expansive soybean dataset, we assessed the performance of these methodologies in selecting features that optimize predictive modeling for various phenotypes. By constructing predictive models based on the selected features, we ascertain the comparative prediction accuracies, thereby illuminating the strengths and limitations of these feature selection methodologies in the realm of genomic data analysis.

Abbreviations: GBLUP, genomic best linear unbiased prediction; GEBV, genomic estimated breeding values; GRM, genomic relationship matrix; GWAS, genome-wide association studies; RF, random forest; SNP, single nucleotide polymorphism; SVM, support vector machine; XGBoost, extreme gradient boosting.

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Plain Language Summary

Modern genetic research relies on vast amounts of data obtained through advanced technologies. This flood of information makes it difficult to find the important genetic markers linked to complex traits. Our study addresses this challenge by exploring different ways to sift through the data to improve the association of genetic markers with traits. We focused on soybean, using a dataset with millions of genetic markers from nearly a thousand different soybean lines. We looked at traditional methods such as genome-wide association studies alongside newer machine learning techniques. By comparing these methods, we aimed to see which ones are best at picking out the most important genetic features for predicting traits. Our findings shed light on the strengths and weaknesses of different feature selection methods in genetic research. This can help researchers choose the best approach for their studies, ultimately improving our understanding of genetic traits.

1 | INTRODUCTION

The advent of high-throughput technologies has facilitated the acquisition of extensive genomic data in plants, holding promise for the discovery of genetic markers and biomarkers related to complex plant traits (Marsh et al., 2021; Sun et al., 2022). Genetic markers have significantly enhanced the efficiency and precision of crop improvement programs through marker-assisted selection (MAS) (Collard & Mackill, 2008). The discovery of quantitative trait loci (QTLs) has enabled the validation of candidate genes associated with specific traits and the application of MAS in crop breeding, marking a pivotal advancement in agricultural genomics (Kushanov et al., 2021). However, the analysis of high-throughput genomic data poses significant challenges due to the high dimensionality and sparsity of the data, which can lead to overfitting, decreased model interpretability, spurious associations, and computational load difficulties (Singh et al., 2015).

High dimensionality refers to the large number of variables or features in the dataset, which can make it computationally challenging and time consuming (Pires & Branco, 2019). For example, the estimated size of the bread wheat (*Triticum aestivum* L.) genome is around 15.5 billion base pairs (Aury et al., 2022), and high-throughput sequencing technologies can generate millions of data points per sample, resulting in datasets with millions of features. Furthermore, many of these features may be redundant or irrelevant, which can make it difficult to extract meaningful information and identify important genetic markers or biomarkers. As a result, dimensionality reduction techniques, such as feature selection or feature extraction, are necessary to reduce the number of variables and improve the efficiency and interpretability of the analysis (Meng et al., 2016; Ray et al., 2021).

Feature selection is an essential step in genomic data analysis. This is because genomic datasets often contain a vast

number of variables or features, many of which may not contribute significantly to the prediction or classification of phenotypes of interest. Also, this can lead to computational challenges when using machine learning models. By selecting a subset of informative features, we can reduce the noise and the dimensionality of the dataset and improve the performance of machine learning models, while also gaining insights into the underlying biological processes that drive the observed data patterns. Moreover, feature selection plays a pivotal role in identifying potential biomarkers associated with economically valuable plant traits, enhancing the accuracy of trait prediction for crop yield (Fei et al., 2022), disease resistance (van Esse et al., 2020), and nutritional quality (Tong & Nikoloski, 2021). As such, the selection of variables stands as an important stage in the analysis of genomic plant data, necessitating the adoption of suitable feature selection techniques to ensure precise and meaningful outcomes in agriculture and economic productivity.

Besides the high dimensionality, sparsity is another common challenge in genomic datasets. Sparsity refers to the fact that many features in genomic datasets have a small or negligible effect on the outcome of interest. For example, in gene expression datasets, most genes are not differentially expressed between groups or conditions of interest. Sparse data can make it difficult to identify meaningful patterns or relationships between features and outcomes, as the noise-to-signal ratio can be high (Atla et al., 2011; Gupta & Gupta, 2019). Additionally, sparsity can lead to model instability, making it challenging to generalize the results to new datasets or populations.

Feature selection is a critical step in addressing the challenges of high dimensionality and sparsity in genomic data analysis and it can help to identify the genetic variants and pathways that are most relevant to a particular trait, disease, or biological process. Feature selection is an important step

in the process of discovering QTLs, as it helps in identifying the most relevant genetic markers that are associated with the traits of interest (Hoff et al., 2019). In the context of genomic selection (GS), the application of feature selection has shown promising results in enhancing predictive performance. Montesinos-Lopez et al. (2023) demonstrated this in their study involving six real datasets, where they found that performing variable selection significantly improved prediction accuracy in four out of the six datasets. This finding confirms the empirical evidence that feature selection can substantially enhance prediction accuracy.

Machine learning methods have emerged as a powerful tool for feature selection in large genomic datasets. It not only allows researchers to identify relevant features and improve the performance of predictive models, but it also helps to interpret their biological relevance in the context of the disease or process of interest (Aono et al., 2020; Gill et al., 2022).

Many existing studies on feature selection for genomic data have limitations, either working with a small number of single nucleotide polymorphism (SNP) markers, which does not reflect the modern genomic data scenarios where millions of SNPs are available for each sample or by analyzing a limited number of samples. Additionally, much of the research has been confined to either human or animal datasets. For instance, Bermingham et al. (2015) assessed the impact of five feature selection methods on prediction using genotype data from 2186 Croatian and 810 UK individuals, housing 263,357 and 260,562 SNPs, respectively. B. Li et al. (2018) leveraged 38,082 SNP markers from 2093 Brahman cattle to evaluate the predictive potential of SNPs selected by random forest (RF), gradient boosting, and extreme gradient boosting (XGBoost). Another study by Y. Li et al. (2022) delved into feature selection for multi-omics cancer data but was restricted to just over 100 individuals. In a study by Aono et al. (2020), a full-sib progeny made up of 219 individuals resulting from a cross between elite sugarcane (*Saccharum officinarum* L.) varieties was utilized. This progeny was characterized by 14,540 SNP markers. The research focused on the selection of features for sugarcane brown rust resistance. Through the testing of five distinct feature selection methods, the team achieved a significant improvement in prediction accuracy. Specifically, they elevated the accuracy from 50% to an impressive 95% using a mere 131 SNP markers, a stark improvement over using the entire set of 14,540 SNPs. Meanwhile, a more recent study by Xiang et al. (2023) compared the feature selection efficacy among various machine learning algorithms, namely, RF, support vector machine (SVM), XGBoost, and convolutional neural network. This comparison was conducted using two distinct pig datasets. The authors utilized pig data genotyped on 42k SNP chips and encompassed seven different pig phenotypes. The sample sizes ranged from 2691 to 6211 individuals, depending on the trait under investigation. The study revealed that feature selection techniques could reduce

Core Ideas

- High-throughput genomic technologies produce vast datasets that reduce the efficiency of trait association.
- Genomic feature selection can increase the accuracy and computational efficiency of trait association.
- Our comparison of feature selection approaches can help readers optimize feature selection and trait association.

the number of genomic markers to 1 in 20, with some traits even exhibiting improved predictive performance compared to utilizing the entire genome dataset.

While feature selection can significantly enhance prediction accuracy in genomic prediction models, several limiting factors must be considered before applying any feature selection method. First, the sample size needs to be sufficiently large to identify features that will generalize well to unseen data. If the training sample size is too small, there may be selection bias, leading to features that do not perform well in predicting new data. Additionally, a potential issue arises when researchers select a discriminative subset of features using the entire dataset and then estimate the prediction model's accuracy with this subset. This approach tests the discrimination rule on samples that were already used to determine the final subset of features, which can lead to an overestimation of the model's performance (Saeys et al., 2007). More importantly, the benefits of feature selection in genomic selection may diminish over time due to various factors. Changes in allele frequencies, driven by selection pressures and genetic drift, can alter the relevance of initially selected features. Recombination events, which occur during meiosis, can break up linked genetic markers, reducing the effectiveness of feature sets that rely on these linkages. Additionally, the inherent instability of the selected features, especially those with smaller effect sizes, can lead to a decline in prediction accuracy as the population evolves.

In our research, we explore further genomic feature selection, comparing the predictive accuracies of features identified by genome-wide association studies (GWAS) with those chosen by two leading machine learning tools, namely, RF (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016). We harnessed a comprehensive soybean dataset encompassing 966 lines with over 5.5 million SNPs (Bayer et al., 2022). This extensive dataset offers a prime opportunity to scrutinize contemporary challenges faced in genomic studies, notably the challenge of dealing with massive feature spaces with a limited sample count (often termed as “small n large p ” dilemmas). Our article begins by offering a comprehensive overview of the various methodologies employed for genomic

feature selection, emphasizing machine learning techniques. Subsequently, using genuine soybean data, we venture into an in-depth analysis, evaluating how well different methods choose important features. By building predictive models based on the selected features, we were able to juxtapose the prediction accuracies for diverse phenotypes, thereby shedding light on the relative power of these feature selection methodologies. Finally, we utilized simulated data to investigate the impact of feature selection over time across future breeding cycles.

2 | MATERIALS AND METHODS

2.1 | Genomic data and phenotypes

The genotype data for this study were sourced from a pool of 1110 soybean lines from the USDA Soybean Germplasm Collection, as previously examined in the soybean pangenome study conducted by Bayer and colleagues in 2022 (Bayer et al., 2022). Briefly, raw sequencing reads underwent a cleaning process where adapter sequences were removed using Trimmomatic v0.36 (Bolger et al., 2014). Following data cleaning, the resulting clean reads were aligned to the reference genome Wm82.a4.v1 (Valliyodan et al., 2019) using BWA-MEM v0.7.17 with default settings (H. Li, 2013). Duplicate reads, which can skew results, were eliminated using Picard tools (<http://broadinstitute.github.io/picard/>). Read realignment was performed using GATK v3.8–1–0 RealignerTargetCreator and IndelRealigner. Subsequently, genetic variants were identified using GATK v3.8–1–0 (McKenna et al., 2010) HaplotypeCaller. Indels and SNPs located on unassembled scaffolds were separated into distinct files to facilitate subsequent analysis. SNPs failing to meet quality criteria (quality by depth < 2.0 || mapping quality < 40.0 || Fisher strand bias > 60.0 || overall quality score < 60.0 || mapping quality rank sum < –12.5 || read position rank sum < –8.0) were removed. To streamline the dataset and focus on informative genetic variations, SNP loci with over 10% missing data were filtered out using VCFTools v3.0 (Danecek et al., 2011). Missing genotypes were inferred, and phasing was conducted using BEAGLE v5.1 (Browning & Browning, 2007; Browning et al., 2018). Data about seed oil content, protein content, and seed weight phenotypes were acquired by downloading information from the USDA-GRIN database (<https://npgsweb.ars-grin.gov/>) for those accessions that possessed accessible observational records.

2.2 | A general overview of feature selection methods

Feature selection methods for genomic prediction can be grouped into three main classes including filter methods,

wrapper methods, and embedded methods (Pudjihartono et al., 2022).

Filter methods utilize statistical measures and tests to rank and select the most relevant genomic features, such as genes or genetic markers, based on their associations with specific biological outcomes or phenotypes. By identifying features that exhibit strong correlations or statistical significance with the target variable, filter methods reduce the dimensionality of genomic data while retaining biologically meaningful information. It evaluates individual feature-outcome relationships using tests such as chi-squared and *t*-tests but may miss redundancy and interactions. For example, in gene expression analysis, filter methods can identify genes that are differentially expressed between experimental conditions or disease states, aiding in the discovery of potential biomarkers or key regulatory genes. One study demonstrating the application of filter methods in genomics is the research by Al-Shahrour et al. (2004), which employed statistical techniques for feature selection in gene expression data analysis. Filter methods are model independent, avoiding bias and overfitting, but may result in a more generalized feature set and potentially lower predictive performance compared to wrapper or embedded methods.

Multivariate techniques such as LASSO, Ridge, and Elastic Net Regression consider multiple features simultaneously. Several studies have demonstrated the effectiveness of LASSO regression in genomic data analysis, including the identification of biomarkers for cancer prognosis and prediction (Bovelstad et al., 2007; Tibshirani, 1997). Furthermore, LASSO regression has been shown to outperform other feature selection techniques, including principal component analysis and univariate feature selection, in terms of both predictive accuracy and model interpretability (Goeman & Bühlmann, 2007; Ogutu et al., 2012). However, LASSO methods might be computationally demanding based on the size of the datasets (Bovelstad et al., 2007).

Wrapper methods for feature selection rely on a chosen model's performance to select the most relevant feature subset, offering higher predictive power compared to filtering methods. They employ heuristic search strategies, such as recursive feature elimination or forward selection, to efficiently explore feature combinations while considering feature interactions. However, wrapper methods are computationally intensive (Y. Li et al., 2022), making them less suitable for high-dimensional data. Additionally, their effectiveness depends on the specific model chosen, and the selected feature subset may not perform well with different models. Despite these limitations, studies such as Choi et al. (2011), Guyon et al. (2002), and Maldonado and Weber (2009) have demonstrated the effectiveness of wrapper methods in improving classification accuracy, making them a valuable tool for feature selection when optimizing model performance is a priority.

Embedded methods seamlessly integrate feature selection into prediction or classifier models. During training, the model optimizes its internal parameters to weigh features and achieve accurate outcomes. This approach combines feature subset search and model construction. Examples include decision tree-based machine-learning algorithms (RF and XGBoost) and regularization models such as LASSO. Embedded methods balance features from filter and wrapper methods, offering computational efficiency and classifier interaction benefits. Several studies have demonstrated the effectiveness of LASSO in genomic data analysis, including the identification of biomarkers for cancer prognosis and prediction (Bovelstad et al., 2007; Tibshirani, 1997). Furthermore, LASSO regression has been shown to outperform other feature selection techniques, including principal component analysis and univariate feature selection, in terms of both predictive accuracy and model interpretability (Goeman & Bühlmann, 2007; Ogutu et al., 2012). However, LASSO might be computationally demanding based on the size of the datasets (Bovelstad et al., 2007).

Some embedded methods support feature interactions, particularly the decision tree-based approaches RF and XGBoost. Díaz-Uriarte and de Andrés (2006) described how the RF algorithm can be efficiently used for gene selection and classification of microarray data. In a study with Brahman cattle genomic data for genomic breeding value prediction, B. Li et al. (2018) found that RF and gradient boosting machine proved effective in selecting SNPs linked to growth-related genes, with their top 3000 SNPs showing similar genomic prediction accuracy to using the entire SNP set and outperformed evenly spaced SNP subsets and XGBoost in terms of accuracy. More recently, Y. Li et al. (2022) found that the permutation importance of RF delivered strong predictive performance when considering only a few selected features.

In the next sub-section, we will provide a brief overview of the RF and XGBoost algorithms.

2.2.1 | Random forest

Leo Breiman first proposed RF algorithms, which are a type of ensemble learning method (Breiman, 2001). Ensemble learning combines multiple models or algorithms to improve prediction or classification performance beyond what could be achieved with individual models. In RF, multiple decision trees are generated based on m random samples selected by bootstrap sampling, where replacement is allowed. To reduce the correlation between features, only a random subset of features is considered at each node of the tree-building process. For classification problems, majority voting is used to predict the class of the validation data, while for regression problems, the mean of the individual tree's prediction is used as the output of the model. The RF algorithm has two key parameters,

the number of decision trees (*num.trees*) and the number of randomly selected features (*mtry*) used to determine the best split at each node. The attraction of the model is based on the following several attributes:

1. a high degree of accuracy;
2. a resistance to overfitting, including in the presence of large numbers of redundant variables;
3. the provision of an out-of-bag error measure; and
4. the provision of a variable ranking at a little extra computational cost.

2.2.2 | Gradient boosting

Gradient boosting are machine learning algorithms that combine multiple weak learning models to create a strong predictor. It works by training a sequence of decision trees, where each new tree attempts to correct the errors made by the previous tree. In other words, the algorithm focuses on the data points that were misclassified by the previous tree and tries to classify them correctly in the next tree. The algorithm uses a loss function to calculate the difference between the predicted and actual values and then updates the model by minimizing this loss function using gradient descent. By iterating this process, the algorithm gradually improves the accuracy of the model until the desired performance is achieved. XGBoost is an advanced implementation of the gradient boosting algorithm that uses a more regularized model formalization to control overfitting and improve generalization performance. XGBoost provides a built-in feature importance ranking that can be used to identify the most relevant features. The algorithm calculates the contribution of each feature in the model based on the number of times it is used to split the data across all trees in the ensemble. The feature importance ranking can be used to select a subset of the most important features for training the model. This can be done by setting a threshold on the importance scores and selecting only the features that exceed the threshold. Alternatively, one can train multiple models using different subsets of features and compare their performance to select the best subset of features.

2.3 | Experimental details

We adopted two distinct approaches to assess prediction accuracies across GWAS, RF, and XGBoost-selected SNPs. In the initial approach, we conducted SNP data filtration solely based on minor allele frequency (MAF), and subsequently, these MAF-filtered datasets were employed for feature ranking in GWAS, RF, and XGBoost. In our second approach, we applied filtration based on MAF and linkage disequilibrium (LD) (at different levels $r^2 > 0.3, 0.5$, and 0.7) to the available

SNP data prior to utilizing them in GWAS, RF, and XGBoost for feature ranking. Furthermore, using simulation data, we examined how the features selected by these three methods hold up across different future generations.

We used PLINK 1.9 (Chang et al., 2015; Purcell et al., 2007) to perform GWAS and LD pruning. To calculate associations between individual SNPs and phenotypes, we utilized the PLINK ‘-assoc’ command. Afterward, we arranged the SNPs according to their *p*-values, assigning higher ranks to those with lower *p*-values as obtained from the PLINK output. For LD pruning, we employed the ‘-indep-pairwise’ command with specific parameters: 100, 10, and 0.3, 0.5, or 0.7. This LD pruning process entailed creating SNP windows of size 100 that did not overlap, shifting the window by 10 SNPs at each iteration, and subsequently removing one SNP from each pair that exceeded pairwise correlation thresholds of 0.3, 0.5, and 0.7, respectively.

To get the feature importance (SNPs ranking) from RF and XGBoost, we used the R packages ranger and xgboost, respectively. ‘ranger’ (Wright & Ziegler, 2017) is a high-performance, scalable, and parallelized implementation for the RF algorithm for classification and regression. We used the importance = ‘impurity’ mode in the ranger package to get the feature ranking. The ‘impurity’ metric corresponds to the Gini index in classification tasks and the variance of the responses in regression scenarios. The R package ‘xgboost’ (Chen & Guestrin, 2016) offers a scalable tree boosting system that not only provides prediction models but also facilitates feature ranking based on model performance. In our analysis, we employed the “Gain” score from the feature importance output of the xgboost model to rank SNPs. The “Gain” score serves as a metric of each feature’s relative contribution to the model’s predictive accuracy, quantifying the enhancement in accuracy or reduction in loss achieved by including a specific feature when making decisions at each split within the XGBoost ensemble’s decision trees.

In our analysis, we applied MTG2 (Lee & van der Werf, 2016) for genomic prediction. Following SNP ranking through GWAS, RF, or XGBoost, MTG2 was utilized to compute genomic estimated breeding values (GEBV). For each specific set of selected SNPs, we initially computed the genomic relationship matrix (GRM) using PLINK. Subsequently, this GRM, in conjunction with phenotype data, was employed to estimate breeding values within the MTG2 framework.

We conducted a comparative analysis of SNP selection methods, including GWAS, RF, and XGBoost, using a 10-fold cross-validation approach. Initially, the available genotype and phenotype data were partitioned into 10 equal folds. One fold served as the validation set, while the remaining data were utilized for ranking SNPs through GWAS, RF, or XGBoost. These SNP rankings were then employed to create diverse SNP subsets for GEBV prediction using MTG2

TABLE 1 genomic best linear unbiased prediction (GBLUP) accuracies were obtained from 10-fold cross-validation using all available single nucleotide polymorphism (SNP) data across three traits: oil content, protein content, and seed weight.

Prediction accuracy	Oil content	Protein content	Seed weight
Mean	0.78	0.54	0.83
Median	0.78	0.55	0.83
Standard deviation	0.03	0.02	0.02
Minimum	0.73	0.52	0.80
Maximum	0.84	0.59	0.86

Note: Prediction accuracies are the Pearson correlation coefficient between predicted and actual phenotypes.

on the validation set. This process was repeated 10 times for each fold, and GEBV prediction accuracy was assessed using the Pearson correlation coefficient between predicted GEBV and actual phenotypes. Ultimately, we averaged the prediction accuracies across the 10 folds to determine the final prediction accuracy for SNP subsets selected by GWAS, RF, and XGBoost.

3 | RESULTS AND DISCUSSION

In this section, we present the outcomes of our benchmark study, which aims to evaluate the predictive accuracy of selected SNPs using three distinct methods: GWAS, RF, and XGBoost. Throughout our analysis, we assessed prediction accuracies by measuring the Pearson correlation coefficient between predicted and actual phenotypes. Table 1 presents the genomic best linear unbiased prediction (GBLUP) accuracies obtained from 10-fold cross-validation using all available SNP data across three traits: oil content, protein content, and seed weight.

Figure 1 describes the GBLUP prediction accuracies for three traits using random sampling across various feature sizes. For each trait, the total available samples were divided into training (80%) and testing (20%) subsets, and SNPs were randomly selected in numbers ranging from 100 to 2,000,000. Prediction accuracies for different feature numbers are the averages from 10-fold cross-validation. The average prediction accuracies with 100 random SNPs were 0.58 for oil content, 0.37 for protein content, and 0.54 for seed weight. As the number of SNPs increased, prediction accuracies gradually improved with some random fluctuations due to random noise. For oil and protein content, prediction accuracies peaked at 0.76 and 0.53, respectively, with around 20,000 SNPs, after which they plateaued. For seed weight, it required about 50,000 SNPs to reach a plateau at 0.82. Table 1 and Figure 1 present benchmarks for prediction accuracies,



FIGURE 1 Trends in prediction accuracy for oil content, protein content, and seed weight as the number of randomly selected single nucleotide polymorphisms (SNPs) increases. Note that the X axis is not scaled.

providing reference points for comparing the performance of three feature selection methods across different scenarios.

3.1 | Prediction accuracies with minimal pre-filtering

The initial genotype dataset encompassed a substantial 5,552,767 variants across 1110 lines. Among these lines, 966 are associated with oil and protein content phenotypes, while 688 lines pertain to seed weight phenotypes. Initially, the available SNP data were subjected to a mild filtration process where SNPs with MAFs less than 0.01 were excluded. Following the MAF filtration ($MAF < 0.01$), the dataset for oil and protein content genotypes was reduced to 5,474,730 SNPs, and the genotype data for seed weight contained 4,944,820 SNPs.

3.1.1 | Prediction accuracy for SNPs selected by GWAS

For each of the phenotypes, we performed a 10-fold GWAS using the MAF-filtered genotype data. In each fold, 80% of the individuals were randomly selected to perform GWAS, while the remaining 20% were kept for genomic prediction based on the n top-ranked SNPs from the GWAS. The prediction accuracy was then calculated as the average prediction accuracy across these 10-fold results. The prediction accuracy was assessed by measuring the correlation coefficient between the actual phenotypes and the predicted GEBV.

For all the phenotypes, we observed a consistent pattern of increased predictive accuracy as the number of SNPs in

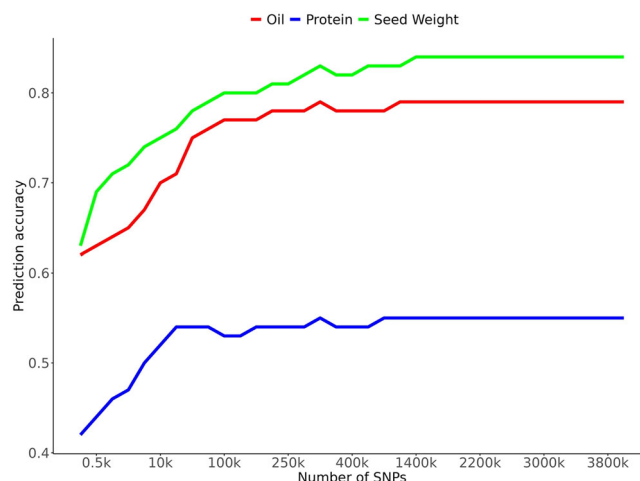


FIGURE 2 Trends in prediction accuracy for oil content, protein content, and seed weight as the number of genome-wide association studies (GWAS) selected single nucleotide polymorphisms (SNPs) increases. Note that the x axis is not scaled.

the dataset increased up to a certain point, beyond which additional variant information did not improve accuracy (Figure 2). Specifically, for the “oil content” phenotype, the prediction accuracy exhibited an upward trend, starting at 0.62 for 100 SNPs and gradually rising to its plateau at 0.79 with 1,000,000 SNPs.

Similarly, the protein content trait displayed a comparable trend. The predictive accuracy increased from 0.42 with 100 SNPs to 0.55 when utilizing 800,000 SNPs. This indicates the utility of a larger dataset for enhancing the precision of the model’s prediction for protein content.

In the case of seed weight, the results consistently demonstrated an improvement in prediction accuracy as the number of SNPs increased. The accuracy scores ranged from 0.63 with 100 SNPs to an impressive 0.84 with 1,400,000 SNPs. Notably, it was apparent that, although there was a point at which prediction accuracies reached their peak, they eventually reached a plateau. This suggests that beyond a certain threshold, additional increases in the number of SNPs do not lead to significant enhancements in predictive power for the phenotypes.

These observations highlight the importance of selecting an optimal number of SNPs for improving predictive models in the context of genetic and genomic research. They also emphasize the need to consider the trade-offs between dataset size and predictive accuracy.

The results of our study reveal a nuanced relationship between the number of SNPs and predictive accuracy. While increasing the number of SNPs leads to improved predictions for oil content, protein content, and seed weight phenotypes, this enhancement is not indefinite. The phenomenon of reaching a plateau in prediction accuracy indicates that there is an

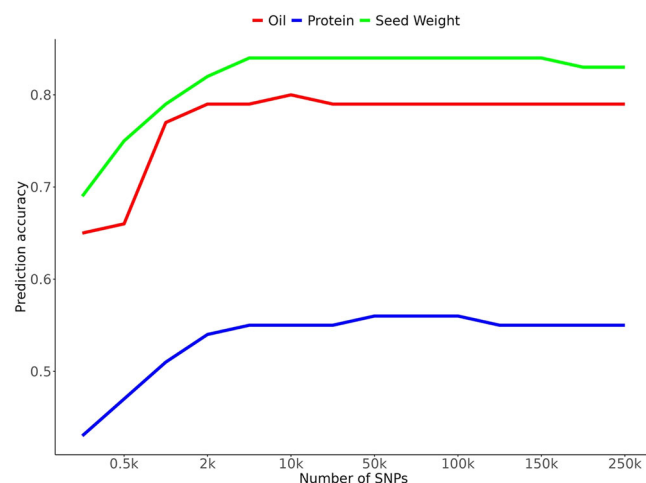


FIGURE 3 Trends in predictive accuracy for oil content, protein content, and seed weight as the number of random forest variable importance selected single nucleotide polymorphisms (SNPs) increases. Note that the x axis is not scaled.

optimal point at which additional SNPs do not significantly contribute to model performance.

These findings have important implications for the design of genetic and genomic studies. Researchers should carefully consider the balance between dataset size and predictive accuracy. Collecting an extensive number of SNPs is valuable, but it may not always lead to substantial improvements in predictive power. Consequently, selecting an optimal subset of informative SNPs can be a more efficient approach.

3.1.2 | Prediction accuracy for SNPs selected by RF variable importance

We performed a 10-fold RF analysis for comparison with the 10-fold GWAS described in the previous section. We used the same 80% of individuals as the previous section for running 10-fold RF and then selected the top-ranked n SNPs from the output of variable importance produced by the RF models. When we examined the variable importance scores produced by the RF models, we found that the variable importance scores are zero beyond 250,000 SNPs for almost all folds. For this reason, we used only SNPs up to 250,000 for genomic prediction.

For each of the phenotypic traits, we evaluated the predictive accuracy while varying the number of SNPs. Figure 3 illustrates the outcomes, displaying the average prediction accuracies for various numbers of SNPs selected by RF variable importance score. For the oil content phenotype, we observed an initial accuracy of 0.65 with 100 SNPs, which consistently improved as more SNPs were incorporated, reaching 0.80 with 10,000 SNPs. Beyond this point, the accuracy remained stable at 0.80 with higher SNP numbers.

The protein content trait demonstrated a similar pattern. Starting at 0.43 with 100 SNPs, the accuracy increased to 0.56 with 100,000 SNPs, after which it plateaued. For the seed weight phenotype, the prediction accuracy began at 0.69 with 100 SNPs and steadily climbed to reach a plateau at 0.84 with 5000 or more SNPs.

These observations align with the trends observed in GWAS-selected SNPs, demonstrating that an optimal number of SNPs significantly enhances predictive accuracy for all three phenotypic traits. Beyond this optimal threshold, the addition of further SNPs does not yield significant improvements in predictive power. However, several notable distinctions exist between these two approaches. Primarily, the utilization of RF variable importance-selected SNPs allows us to attain comparable levels of accuracy as GWAS but with a significantly reduced number of SNPs. When employing RF feature selection, the peak accuracy for oil content, protein content, and seed weight stood at 0.80, 0.56, and 0.84, and was achieved with just 10,000, 50,000, and 2,000 SNPs, respectively. In contrast, with GWAS, the peak accuracies were 0.79, 0.55, and 0.84, and these were achieved using 1,200,000, 800,000, and 1,400,000 SNPs, respectively. The second distinctive feature lies in the post-peak accuracy behavior. In RF, there was a slight decrease in accuracy after reaching the peak before it eventually plateaued. In contrast, in GWAS, the prediction accuracies remained stable once the peak was attained.

The difference in the number of SNPs required to attain similar levels of accuracy between GWAS-selected SNPs and those chosen based on RF variable importance can be elucidated by multiple factors. Since we applied minimal filtration to the genotype data, except for MAF, the dataset contains numerous correlated SNPs. In GWAS, which concentrates on individual SNP-trait associations, these correlated SNPs tend to rank similarly. Consequently, when selecting a subset of SNPs from the top-ranked GWAS list, many of these SNPs may be correlated and point to the same QTL region. Consequently, choosing a limited number of top-ranked GWAS SNPs may not fully capture the entirety of genomic variation within the dataset. As the number of SNPs increases, the dataset's variation is more comprehensively captured, leading to improved prediction accuracy.

In contrast, RF employs a distinct feature importance metric based on the reduction of impurity in decision tree nodes. This metric identifies SNPs that wield a substantial influence on the model's performance. RF prioritizes SNPs that make the most significant contributions to predictive power, resulting in a more concise and informative subset. Moreover, RF is designed to detect interactions between SNPs and other features (Wright et al., 2016), enabling it to identify combinations of SNPs that collectively exert a potent influence on predictions. This capacity allows RF to achieve comparable accuracy with a smaller subset of SNPs.

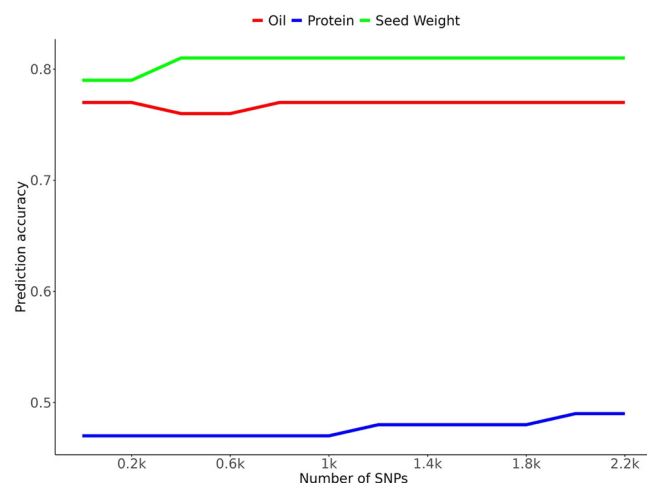


FIGURE 4 Trends in predictive accuracy for oil content, protein content, and seed weight as the number of extreme gradient boosting (XGBoost) variable importance selected single nucleotide polymorphisms (SNPs) increases. Note that the x axis is not scaled.

3.1.3 | Prediction accuracy for SNPs selected by XGBoost feature importance

In alignment with our previous methods for GWAS and RF feature selection, we performed a 10-fold XGBoost analysis on the genotype data to assess feature importance. We maintained consistency by utilizing the same set of individuals for training the XGBoost models, which enabled us to determine the top-ranked SNPs. Subsequently, we executed genomic prediction on the remaining 20% of the data. It is important to note that XGBoost models generate feature importance scores exclusively for the features employed in constructing the decision trees. Consequently, the quantity of SNPs for which we obtained feature importance scores via XGBoost was limited. In our comprehensive 10-fold XGBoost analysis, the count of SNPs featuring these importance scores ranged between 2200 and 2300. Due to this constraint, we opted to incorporate only those SNPs that were ranked within the top 2200 by XGBoost's feature ranking methodology for our genomic prediction benchmark.

Consistent with our previous approach, we conducted an evaluation of predictive accuracy for each of the three traits by varying the number of SNPs selected via XGBoost. As shown in Figure 4, the chart illustrates the mean prediction accuracies across different quantities of SNPs chosen based on XGB feature importance scores. For the oil content phenotype, we observed an initial accuracy of 0.77 with a mere 100 SNPs, experienced a minor decline at 400 and 600 SNPs, followed by a resurgence at 1000 SNPs, and eventually reached a plateau, maintaining consistent accuracy levels up to 2200 SNPs. In the case of the protein content trait, the prediction accuracy initiated at 0.43 with 100 SNPs. Subsequently, the accuracy exhibited an upward trend, reaching 0.48 with 1200

SNPs and further elevating to 0.49 with 2000 SNPs. For the seed weight phenotype, the prediction accuracy embarked at a substantial 0.79 with 100 SNPs. It ascended to 0.81 with 400 SNPs and remained steadfast, indicative of a plateau in predictive accuracy.

3.1.4 | Prediction accuracy comparison between SNPs selected by three methods

In Figures 5–7, we present a detailed comparison of the prediction performance associated with SNPs chosen by three different methods: GWAS, RF, and XGBoost for the traits: oil content, protein content, and seed weight. To provide a comprehensive understanding, we have illustrated the prediction accuracies using boxplots based on a 10-fold cross-validation. This is done while incrementally increasing the number of top-ranked SNPs, starting from an initial count of 100 and going up to a maximum of 2200. It is essential to note that, to maintain an equitable comparison, we have limited our evaluation to a maximum of 2200 top SNPs. This choice was made because XGBoost typically yields between 2200 and 2300 SNPs across its different model outputs.

A few observations can be derived from Figures 5–7 as follows:

- Overall performance:** Across all three trait evaluations, the SNPs selected via XGBoost consistently outperformed those selected by both GWAS and RF. This observation aligns with recent research by Xiang et al. (2023) conducted with pig genotypes. In their genomic feature selection experiment, the most stable and accurate outcomes across various algorithms were achieved by employing XGBoost feature selection in conjunction with the SVM prediction algorithm. Among the three methods examined, SNPs identified via GWAS exhibited the least effective prediction accuracy.
- Trend with increasing SNPs:** As we expand the number of SNPs under consideration, there is a noticeable enhancement in the performance of the RF-selected SNPs. Intriguingly, when the count crosses roughly 1200 SNPs, RF's performance surpasses that of XGBoost. However, despite increasing the number of SNPs, the GWAS method lags behind, showing subpar performance compared to both RF and XGBoost.
- Variation in accuracies between folds:** The fluctuation in prediction accuracies among different folds is notably more pronounced for SNPs selected by GWAS compared to the other two methods. This suggests that the GWAS method may be picking up SNPs in each fold that amount to random noise rather than genuinely contributing to trait prediction.

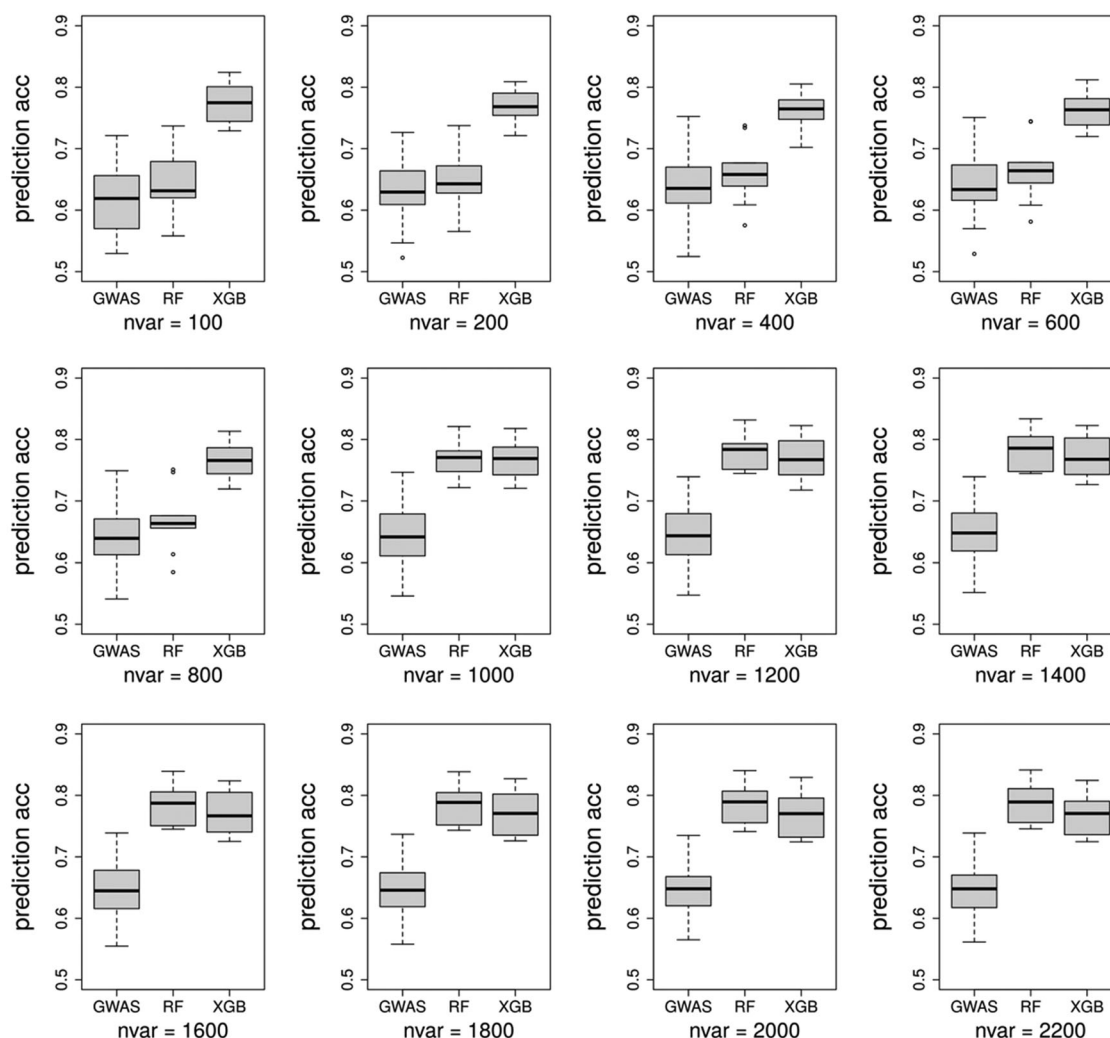


FIGURE 5 Prediction performance comparison for oil content phenotype between SNPs selected by genome-wide association studies (GWAS), random forest, and extreme gradient boosting (XGBoost). Prediction acc refers to the Pearson correlation between the actual and predicted values; nvar denotes the number of single nucleotide polymorphisms used to train the prediction model.

3.2 | Prediction accuracy with LD pruning and MAF-filtering

In the first part of the results section, we discussed the prediction accuracy for SNPs chosen using GWAS, RF, and XGBoost, applying only a basic filter based on MAF ($MAF > 0.01$). This limited filtering allowed us to utilize the most extensive set of SNPs for the RF, XGBoost, and GWAS techniques. Given that our starting dataset has over 5.5 million SNPs, many of them are closely interrelated due to high LD. Such strong interrelations between features can pose challenges as they might influence the predictive performance of models, though the impact varies across different model types. Correlated features can induce multicollinearity, complicating the understanding of each feature's predictive significance (Lindner et al., 2022). Moreover, these overlapping features risk overfitting, making the model overly adapted to training data and less effective on unseen data. To address the chal-

lenges posed by correlated features, we will further detail the prediction performance of SNPs chosen by the three methods after filtering based on LD in the subsequent sections of our results. We conducted experiments using three distinct LD (r^2) thresholds: 0.3, 0.5, and 0.7. After applying the chosen LD threshold to prune the initial dataset, we fed the pruned SNPs into the RF, XGBoost, or GWAS methods for ranking. Subsequently, we employed the ranked SNPs (using various SNP counts) for predictions via GBLUP (Clark & van der Werf, 2013).

3.2.1 | Prediction comparisons between LD-filtered SNPs and SNPs without LD-filtering

Figures 8–10 display a comparison between the prediction performances using LD-pruned SNPs and those without LD pruning. Figures 8a–c outline the outcomes for oil content,

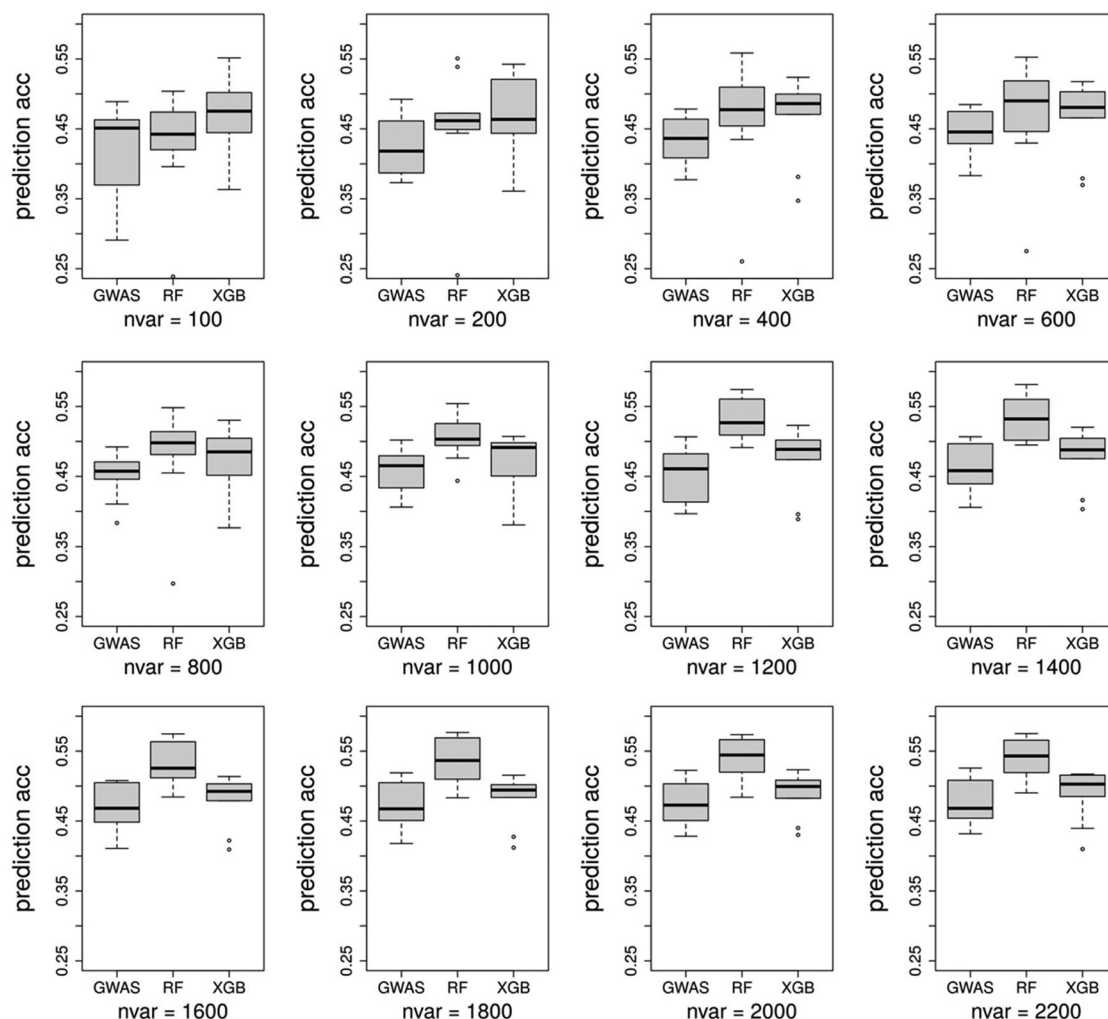


FIGURE 6 Prediction performance comparison for protein content phenotype between single nucleotide polymorphisms (SNPs) selected by genome-wide association studies (GWAS), random forest, and extreme gradient boosting (XGBoost). Prediction acc refers to the Pearson correlation between the actual and predicted values; nvar denotes the number of SNPs used to train the prediction model.

protein content, and seed weight using GWAS-selected SNPs, both pre- and post-LD-pruning at various thresholds. The black line serves as a benchmark, illustrating results without any LD pruning before the GWAS. The blue, green, and red lines, respectively, depict outcomes for LD thresholds of < 0.3 , < 0.5 , and < 0.7 .

From Figure 8a–c, it is clear that pre-filtering based on LD improves accuracy. The initial observation is that when SNPs are filtered based on their LD before conducting a GWAS, the prediction accuracy for the three phenotypes improves. This suggests that removing SNPs that are in high LD (i.e., closely linked and tend to be inherited together) can help in getting a clearer signal in the data, as such SNPs might introduce noise or redundancy. Also, a strict LD threshold works best for fewer SNPs. The strictest LD threshold of 0.3 provides the highest accuracy when the number of SNPs used in the prediction is low. A possible reason might be that, with fewer SNPs, it is essential to ensure that each SNP provides unique information. By setting a stringent LD threshold, one

ensures that the selected SNPs are mostly independent and non-redundant, thus maximizing the information gained from a limited set of SNPs. However, as the SNP count increases, the LD threshold impact diminishes. As more SNPs are added to the analysis, the differences in prediction accuracy among the three LD thresholds become less pronounced. By the point 200k SNPs are being considered, the prediction accuracies across the different thresholds are nearly identical. This could be because, with a large number of SNPs, the dataset captures a broader spectrum of genetic variation. As a result, the specific LD pruning threshold becomes less critical. The sheer amount of data might compensate for any noise or redundancy introduced by SNPs that are in LD.

Figures 9a–c depict the results for the three phenotypes when using SNPs selected by RF, before and after applying LD-pruning at different thresholds. When there are fewer SNPs to work with, removing those that are highly correlated (via LD pre-filtering) before using them in the RF model boosts the prediction accuracy. This is similar to what was

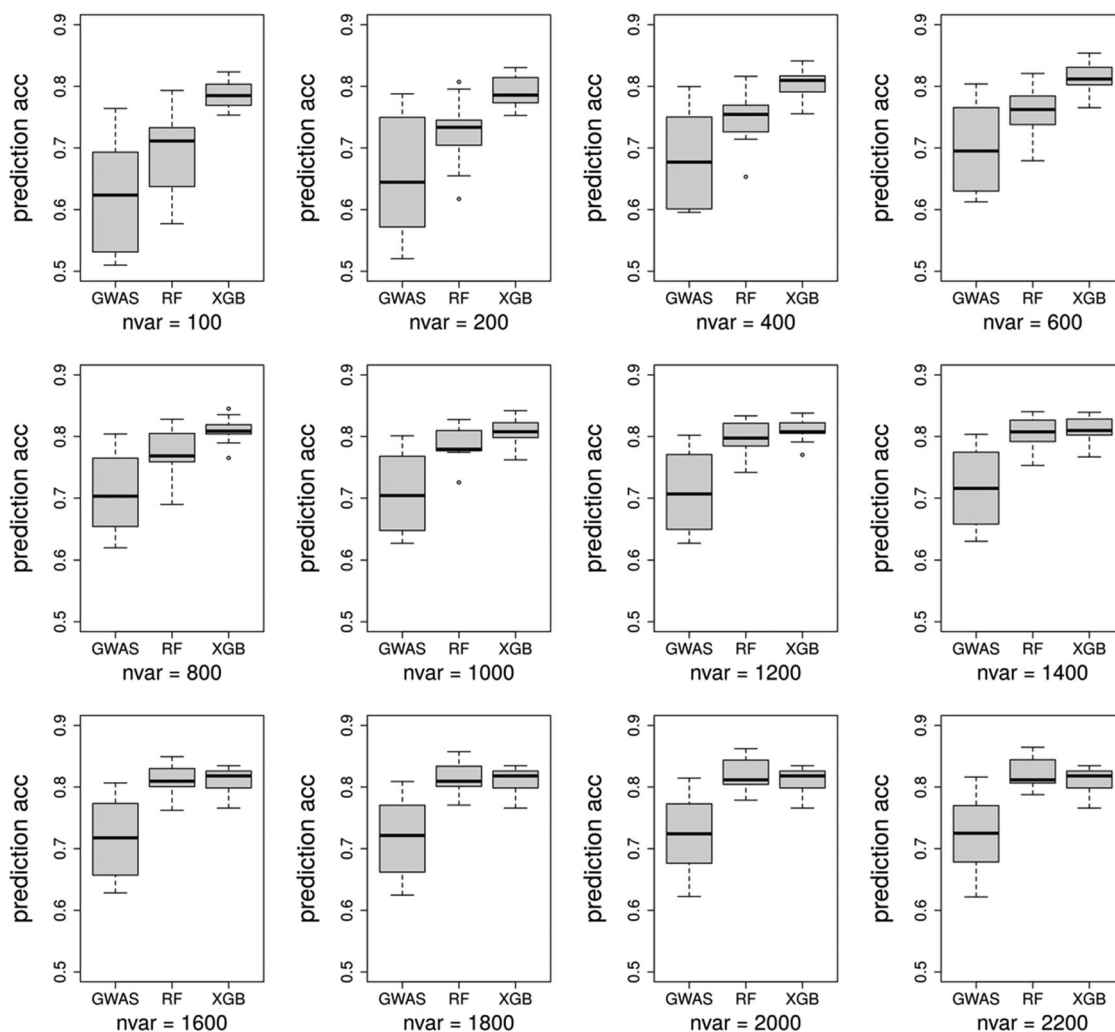


FIGURE 7 Prediction performance comparison for seed weight phenotype between single nucleotide polymorphisms (SNPs) selected by genome-wide association studies (GWAS), random forest, and extreme gradient boosting (XGBoost). Prediction acc refers to the Pearson correlation between the actual and predicted values; nvar denotes the number of SNPs used to train the prediction model.

observed with the GWAS method. It suggests that, for smaller SNP sets, reducing redundancy and potential noise by pruning based on LD is beneficial. As the number of SNPs increases, the distinction in prediction accuracy between using pre- and post-LD-pruned SNPs becomes less evident. This contrasts with the GWAS method, where LD pruning consistently made a more pronounced difference in accuracy. In some cases, the RF algorithm even performed slightly better without LD pruning. This characteristic highlights one of the strengths of the RF algorithm: its ability to handle and manage noisy data. Essentially, while LD pruning can help reduce noise by removing correlated SNPs, the RF algorithm is already well-equipped to handle such noise due to its ensemble nature and decision tree structure.

Figures 10a–c show the prediction accuracies for three phenotypes using SNPs selected by the XGBoost algorithm before and after LD pruning. From an examination of these figures, it emerges that the process of LD pruning does not seem to favor the prediction outcomes for SNPs when they

are selected through the XGBoost algorithm. Specifically, for oil content and seed weight phenotypes, the data indicates a more promising prediction performance when the XGBoost algorithm uses SNPs that have not undergone any LD pruning. This contrasts with the GWAS and RF methods, where LD pruning helped in filtering out noise and improved the prediction performance of the selected SNPs. This suggests that for these two traits, the raw, unpruned data might provide richer or more diverse information for the XGBoost model to leverage. The protein content trait presents a slightly different scenario. When the number of SNPs ranged between 400 and 2000, the SNPs that had not been subjected to LD pruning exhibited slightly diminished prediction performance compared to the SNPs that had been pruned at an LD threshold of 0.5. However, an interesting twist appears when the number of SNPs surpasses 2000: in this case, the prediction performance derived from the non-LD-pruned SNPs overtakes that of the predictions made from the LD-pruned SNPs.

FIGURE 8 Prediction performance comparison for phenotypes: (a) oil content, (b) protein content, and (c) seed weight using single nucleotide polymorphisms (SNPs) selected by genome-wide association studies (GWAS), both before and after linkage disequilibrium (LD) pruning. The black line depicts prediction accuracies without any LD adjustments. The blue line indicates results when SNPs with LD (r^2) > 0.3 within a 100 kb window were excluded. Meanwhile, the red and green lines showcase outcomes after applying LD pruning at thresholds of 0.5 and 0.7, respectively.

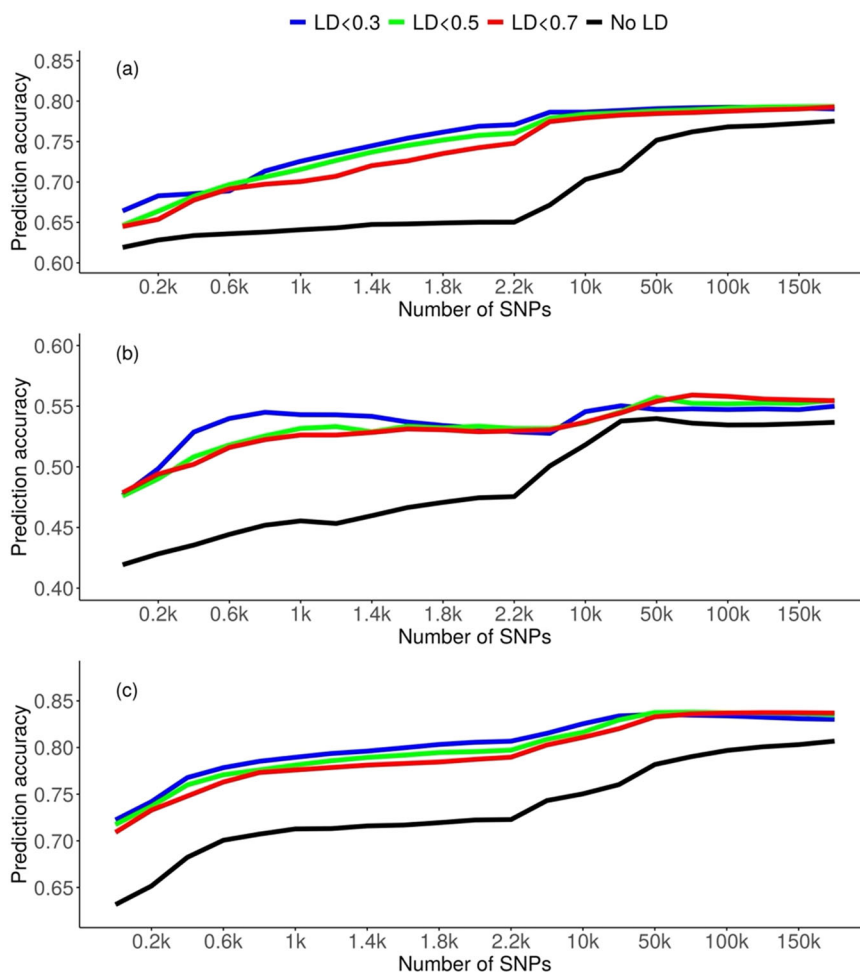


TABLE 2 Comparison of genomic best linear unbiased prediction (GBLUP) prediction performance for oil content, protein content, and seed weight after linkage disequilibrium (LD) pruning at three different thresholds, with no additional feature selection applied.

LD threshold	Oil content	Protein content	Seed weight
0.3	0.789	0.552	0.828
0.5	0.790	0.558	0.830
0.7	0.791	0.558	0.832

3.2.2 | Prediction accuracy comparison with LD pruned SNPs selected by three methods

We conducted a comparison of prediction accuracies using top SNPs selected by GWAS, RF, and XGBoost after LD pruning at thresholds of 0.3, 0.5, and 0.7. Table 2 presents prediction accuracies using LD-pruned data without any additional feature selection. Tables 3–5 show the prediction accuracies for the oil content, protein content, and seed weight phenotypes, respectively, with different numbers of features selected after LD pruning.

LD pruning notably enhanced the predictive capacity of the GWAS-selected method. Nonetheless, employing a too-

lenient threshold (i.e., $r^2 > 0.7$) might still introduce noise into the data, potentially diminishing the predictive efficacy of the chosen SNPs. Conversely, an excessively stringent LD threshold (i.e., $r^2 > 0.3$) could eliminate SNPs that are important for the prediction model. Our findings highlight that the $r^2 > 0.5$ threshold strikes a balance by effectively removing noisy or redundant SNPs while retaining those valuable for the predictive model. Except for a single scenario ($r^2 > 0.3$ for protein content), RF consistently outperformed GWAS and XGBoost-selected SNPs in terms of prediction accuracy. Both RF and XGBoost displayed resilience to LD filtering thresholds, highlighting their ability to handle datasets with abundant redundant features and noise. Despite being the least performing, XGBoost-selected SNPs exhibited the highest consistency across different SNP quantities.

Our study sheds light on important aspects when working with SNP data that boasts an extensive array of features. When leveraging GWAS for feature selection within such datasets, the implementation of LD pruning or similar techniques improves prediction accuracy. This arises from a recurring issue: closely correlated features in SNP data tend to acquire similar rankings during GWAS analyses. As a consequence, when SNPs are cherry-picked based solely on GWAS-derived

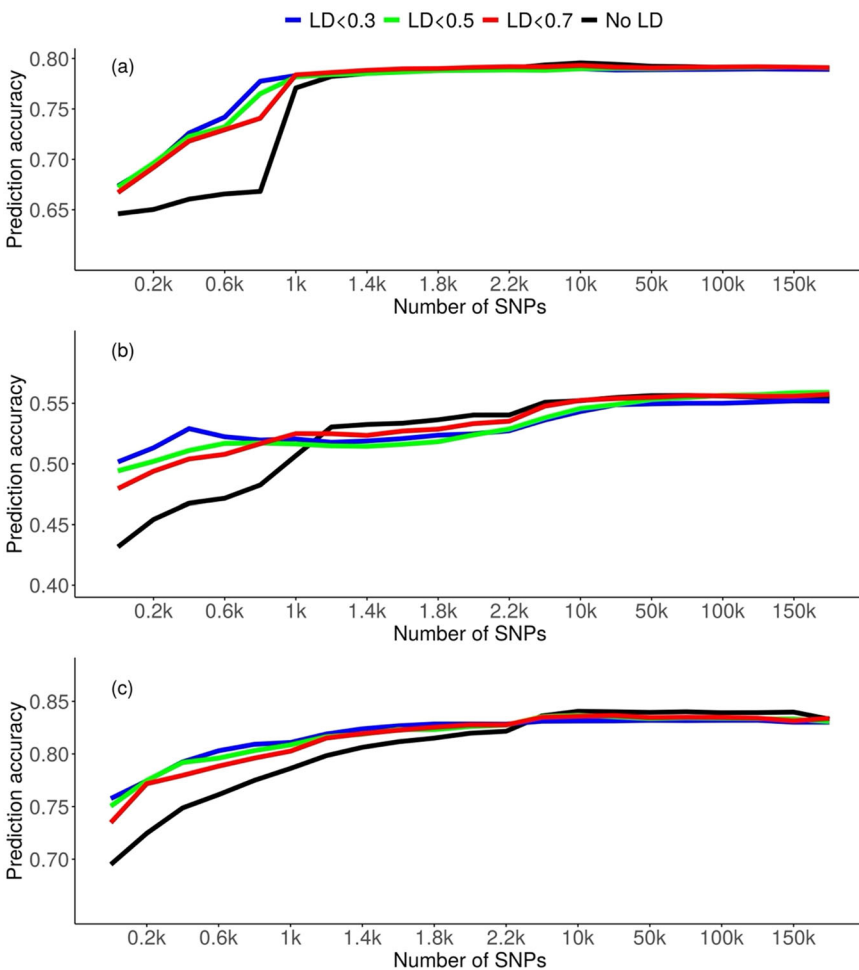


FIGURE 9 Prediction performance comparison for phenotypes: (a) oil content, (b) protein content, and (c) seed weight using single nucleotide polymorphisms (SNPs) selected by random forest, both before and after linkage disequilibrium (LD) pruning. The black line depicts prediction accuracies without any LD adjustments. The blue line indicates results when SNPs with LD (r^2) greater than 0.3 within a 100 kb window were excluded. Meanwhile, the red and green lines showcase outcomes after applying LD pruning at thresholds of 0.5 and 0.7, respectively.

TABLE 3 Prediction performance comparison for oil content phenotype between single nucleotide polymorphisms (SNPs) after linkage disequilibrium (LD) pruning at three different thresholds and then selected by genome-wide association studies (GWAS), random forest (RF), and extreme gradient boosting (XGBoost).

Number of SNPs	LD (r^2) > 0.3			LD (r^2) > 0.5			LD (r^2) > 0.7		
	GWAS	RF	XGBoost	GWAS	RF	XGBoost	GWAS	RF	XGBoost
100	0.664	0.674	0.769	0.646	0.672	0.766	0.645	0.667	0.766
200	0.683	0.694	0.766	0.664	0.696	0.766	0.654	0.692	0.764
400	0.685	0.726	0.765	0.682	0.723	0.765	0.678	0.718	0.761
600	0.689	0.742	0.765	0.697	0.732	0.765	0.691	0.729	0.760
800	0.714	0.777	0.765	0.706	0.765	0.763	0.697	0.741	0.765
1000	0.726	0.783	0.763	0.715	0.782	0.761	0.700	0.784	0.762
1200	0.735	0.786	0.765	0.727	0.784	0.762	0.707	0.786	0.759
1400	0.745	0.787	0.763	0.737	0.785	0.759	0.720	0.788	0.761
1600	0.754	0.788	0.763	0.745	0.787	0.757	0.726	0.790	0.762
1800	0.762	0.789	0.766	0.752	0.788	0.758	0.735	0.790	0.763
2000	0.769	0.790	0.766	0.758	0.788	0.759	0.742	0.791	0.763
2200	0.771	0.790	0.767	0.760	0.789	0.761	0.748	0.792	0.764

Note: The bold values signify the highest prediction accuracy achieved among the three methods for a specific LD threshold.

FIGURE 10 Prediction performance comparison for phenotypes: (a) oil content, (b) protein content, and (c) seed weight using single nucleotide polymorphisms (SNPs) selected by extreme gradient boosting (XGBoost), both before and after linkage disequilibrium (LD) pruning. The black line depicts prediction accuracies without any LD adjustments. The blue line indicates results when SNPs with LD (r^2) greater than 0.3 within a 100 kb window were excluded. Meanwhile, the red and green lines showcase outcomes after applying LD pruning at thresholds of 0.5 and 0.7, respectively.

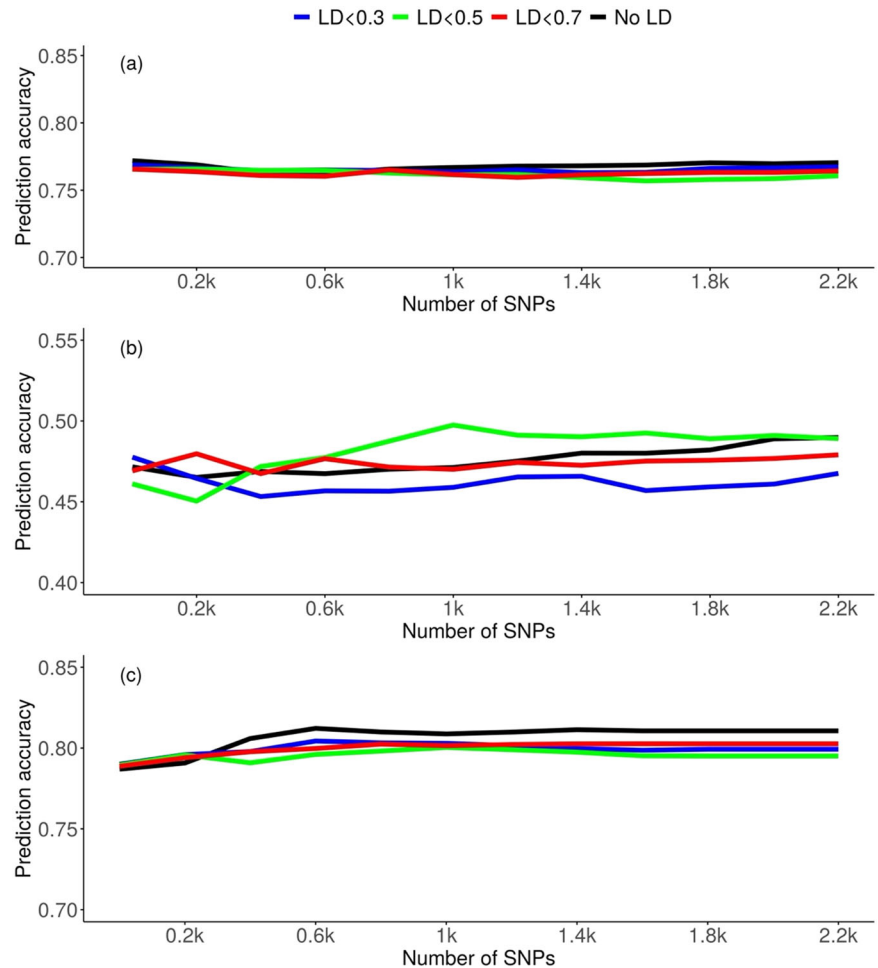


TABLE 4 Prediction performance comparison for protein content phenotype between single nucleotide polymorphisms (SNPs) after linkage disequilibrium (LD) pruning at three different thresholds and then selected by genome-wide association studies (GWAS), random forest (RF), and extreme gradient boosting (XGBoost).

Number of SNPs	LD (r^2) > 0.3			LD (r^2) > 0.5			LD (r^2) > 0.7		
	GWAS	RF	XGBoost	GWAS	RF	XGBoost	GWAS	RF	XGBoost
100	0.476	0.502	0.478	0.476	0.494	0.461	0.479	0.479	0.469
200	0.498	0.513	0.464	0.490	0.502	0.450	0.494	0.494	0.480
400	0.529	0.529	0.453	0.508	0.511	0.472	0.502	0.504	0.467
600	0.540	0.522	0.457	0.518	0.517	0.477	0.516	0.508	0.477
800	0.545	0.520	0.457	0.525	0.517	0.487	0.522	0.517	0.471
1000	0.543	0.520	0.459	0.532	0.516	0.497	0.526	0.525	0.470
1200	0.543	0.518	0.465	0.533	0.515	0.491	0.526	0.525	0.474
1400	0.542	0.519	0.466	0.529	0.514	0.490	0.528	0.523	0.473
1600	0.537	0.521	0.457	0.533	0.516	0.492	0.531	0.527	0.475
1800	0.534	0.524	0.459	0.532	0.518	0.489	0.531	0.529	0.476
2000	0.532	0.525	0.461	0.533	0.524	0.491	0.529	0.533	0.477
2200	0.529	0.527	0.468	0.532	0.529	0.489	0.530	0.535	0.479

Note: The bold values signify the highest prediction accuracy achieved among the three methods for a specific LD threshold.

TABLE 5 Prediction performance comparison for seed weight phenotype between single nucleotide polymorphisms (SNPs) after linkage disequilibrium (LD) pruning at three different thresholds and then selected by genome-wide association studies (GWAS), random forest (RF), and extreme gradient boosting (XGBoost).

Number of SNPs	LD (r^2) > 0.3			LD (r^2) > 0.5			LD (r^2) > 0.7		
	GWAS	RF	XGBoost	GWAS	RF	XGBoost	GWAS	RF	XGBoost
100	0.722	0.758	0.790	0.717	0.750	0.789	0.709	0.735	0.789
200	0.742	0.774	0.796	0.737	0.775	0.796	0.733	0.772	0.794
400	0.768	0.792	0.798	0.760	0.792	0.791	0.748	0.780	0.798
600	0.778	0.803	0.804	0.771	0.796	0.796	0.763	0.788	0.800
800	0.785	0.809	0.803	0.776	0.803	0.798	0.773	0.796	0.802
1000	0.789	0.811	0.803	0.781	0.808	0.800	0.776	0.803	0.801
1200	0.794	0.819	0.802	0.786	0.816	0.799	0.779	0.815	0.802
1400	0.796	0.824	0.800	0.789	0.819	0.797	0.781	0.819	0.803
1600	0.800	0.827	0.799	0.792	0.823	0.795	0.783	0.822	0.803
1800	0.803	0.828	0.799	0.795	0.823	0.795	0.784	0.825	0.803
2000	0.806	0.828	0.799	0.795	0.826	0.795	0.787	0.827	0.803
2200	0.807	0.828	0.799	0.797	0.827	0.795	0.790	0.827	0.803

Note: The bold values signify the highest prediction accuracy achieved among the three methods for a specific LD threshold.

p -values, they might inadequately represent the full spectrum of genetic variation due to this inherent redundancy.

However, within this complex landscape, both RF and XGBoost demonstrate remarkable proficiency in feature selection, even when confronted with highly correlated data. Interestingly, the removal of highly correlated features before the selection process negatively impacts XGBoost's efficacy in choosing SNPs, highlighting the intricacies of its feature selection mechanism. In our study, we found that machine learning algorithms possess an inherent adaptability to handle the challenges posed by datasets containing high redundancy and noise levels. Their capability to navigate through these complexities underscores their reliability in extracting meaningful information from genetic datasets, regardless of the presence of redundant features or noise. Of the three feature selection methods evaluated using extensive genomic data, RF emerged as the most robust and dependable. However, for the design of a cost-effective SNP chip containing a limited number of SNPs, XGBoost proves to be a viable choice, offering good predictive performance.

3.3 | Prediction accuracy over the breeding cycles

Sections 3.1 and 3.2 compare prediction accuracies within a population where the available datasets were divided into training (for feature selection) and testing (for genomic prediction using selected features). In this setup, haplotypes were static in terms of size and frequency. However, in real-world breeding cycles, allele frequencies change over time, and recombination breaks down larger haplotypes into smaller ones, some of which may segregate without any marker tag-

ging them. Consequently, feature selection could become a constraint for future cycles. In this section, we explore the effect of feature selection over different breeding cycles.

We simulated 10 generations using the R package “epinetR” (Detterer et al., 2024), utilizing the actual genotypes from 1110 soybean lines as the founder population. In our simulation, we used a heritability of 0.46, consistent with the heritability of protein content in the actual dataset. Approximately 100 SNPs were selected from each of the 20 chromosomes as the QTL. The phenotypes were generated under an additive model as the sum of the QTL effects plus random noise. We generated about 300 individuals per generation from the founder population.

For each generation of the simulated genotypes and phenotypes, we randomly selected 220 individuals and predicted the GEBV using features selected from the founder populations with GWAS, RF, and XGBoost. This was conducted in 10 folds. Tables 6–8 present the mean prediction accuracies across different generations from generation 2 to generation 11, with different numbers of features selected by GWAS, RF, and XGBoost, respectively.

The results from our study indicate a clear trend in the prediction accuracies of GEBV across multiple generations, as demonstrated in Tables 6–8. Utilizing features selected by GWAS, RF, and XGBoost, we observed that prediction accuracies generally decreased over generations, with notable random fluctuations likely attributed to random noise. Particularly, with a smaller number of features (e.g., 500), the decline in prediction accuracies was pronounced across generations. However, when a larger number of features were employed, the prediction accuracies remained relatively high for the initial few generations (up to the third or fourth) before gradually deteriorating. This suggests that while using

TABLE 6 Mean prediction accuracies for single nucleotide polymorphisms (SNPs) selected by genome-wide association studies (GWAS) from the founder population across different simulated generations with varying numbers of SNPs.

Generation	500 SNPs	1000 SNPs	2000 SNPs	10,000 SNPs	20,000 SNPs	50,000 SNPs
2	0.275 (0.03)	0.296 (0.03)	0.304 (0.01)	0.31 (0.03)	0.315 (0.03)	0.318 (0.03)
3	0.268 (0.02)	0.281 (0.02)	0.314 (0.03)	0.319 (0.03)	0.329 (0.03)	0.325 (0.03)
4	0.261 (0.05)	0.291 (0.03)	0.289 (0.03)	0.333 (0.02)	0.332 (0.03)	0.338 (0.03)
5	0.165 (0.04)	0.195 (0.03)	0.218 (0.04)	0.247 (0.01)	0.256 (0.01)	0.261 (0.01)
6	0.218 (0.01)	0.265 (0.02)	0.276 (0.04)	0.289 (0.01)	0.299 (0.01)	0.293 (0.01)
7	0.174 (0.03)	0.212 (0.02)	0.263 (0.03)	0.311 (0.03)	0.323 (0.02)	0.332 (0.01)
8	0.109 (0.04)	0.162 (0.05)	0.165 (0.07)	0.187 (0.04)	0.183 (0.02)	0.183 (0.02)
9	0.162 (0.05)	0.205 (0.05)	0.248 (0.04)	0.314 (0.02)	0.336 (0.02)	0.35 (0.02)
10	0.04 (0.05)	0.047 (0.04)	0.089 (0.03)	0.197 (0.02)	0.247 (0.01)	0.275 (0.01)
11	0.142 (0.04)	0.141 (0.05)	0.200 (0.04)	0.294(0.03)	0.316 (0.02)	0.315 (0.01)

Note: The numbers in parentheses represent the standard error.

TABLE 7 Mean prediction accuracies for single nucleotide polymorphisms (SNPs) selected by random forest from the founder population across different simulated generations with varying numbers of SNPs.

Generation	500 SNPs	1000 SNPs	2000 SNPs	10,000 SNPs	20,000 SNPs	50,000 SNPs
2	0.352 (0.05)	0.354 (0.03)	0.35 (0.031)	0.341 (0.02)	0.341 (0.03)	0.34 (0.02)
3	0.32 (0.03)	0.324 (0.03)	0.337 (0.03)	0.331 (0.04)	0.33 (0.04)	0.337 (0.03)
4	0.239 (0.04)	0.263 (0.04)	0.283 (0.05)	0.317 (0.03)	0.324 (0.04)	0.339 (0.04)
5	0.149 (0.04)	0.158 (0.02)	0.189 (0.03)	0.212 (0.02)	0.232 (0.02)	0.25 (0.02)
6	0.181 (0.04)	0.214 (0.05)	0.213 (0.07)	0.261 (0.04)	0.274 (0.04)	0.283 (0.03)
7	0.124 (0.04)	0.125 (0.05)	0.14 (0.03)	0.201 (0.05)	0.211 (0.04)	0.212 (0.03)
8	0.126 (0.05)	0.116 (0.06)	0.104 (0.09)	0.139 (0.05)	0.14 (0.04)	0.137 (0.05)
9	0.066 (0.07)	0.082 (0.05)	0.12 (0.03)	0.164 (0.04)	0.156 (0.03)	0.149 (0.03)
10	0.01 (0.06)	0.031 (0.04)	0.1 (0.02)	0.155 (0.05)	0.172 (0.06)	0.172 (0.03)
11	0.047 (0.05)	0.049 (0.06)	0.096 (0.03)	0.096 (0.02)	0.091 (0.05)	0.134 (0.03)

Note: The numbers in parentheses represent the standard error.

TABLE 8 Mean prediction accuracies for single nucleotide polymorphisms (SNPs) selected by XGBoost from the founder population across different simulated generations with varying numbers of SNPs.

Generation	500 SNPs	1000 SNPs	2000 SNPs	2500 SNPs	3000 SNPs	3500 SNPs
2	0.326 (0.05)	0.333 (0.04)	0.342 (0.04)	0.34 (0.03)	0.327 (0.03)	0.329 (0.03)
3	0.307 (0.05)	0.333 (0.05)	0.321 (0.04)	0.312 (0.03)	0.302 (0.04)	0.298 (0.04)
4	0.238 (0.04)	0.270 (0.04)	0.275 (0.03)	0.282 (0.04)	0.284 (0.03)	0.287 (0.03)
5	0.133 (0.03)	0.159 (0.05)	0.156 (0.06)	0.162 (0.06)	0.171 (0.06)	0.179 (0.07)
6	0.147 (0.05)	0.157 (0.03)	0.175 (0.03)	0.190 (0.03)	0.201 (0.01)	0.203 (0.02)
7	0.128 (0.07)	0.183 (0.05)	0.209 (0.08)	0.200 (0.06)	0.175 (0.04)	0.186 (0.05)
8	0.105 (0.02)	0.097 (0.03)	0.117 (0.04)	0.119 (0.04)	0.103 (0.03)	0.095 (0.03)
9	0.030 (0.06)	0.043 (0.10)	0.016 (0.12)	0.016 (0.08)	0.023 (0.06)	0.024 (0.05)
10	0.055 (0.05)	0.033 (0.05)	0.099 (0.05)	0.109 (0.06)	0.115 (0.05)	0.125 (0.06)
11	0.022 (0.04)	0.020 (0.08)	0.030 (0.08)	0.032 (0.07)	0.046 (0.07)	0.055 (0.06)

Note: The numbers in parentheses represent the standard error.

a more extensive feature set can temporarily sustain prediction accuracy, the inherent changes in allele frequencies and recombination over time still ultimately impact the effectiveness of genomic prediction models.

Our study demonstrates that prediction accuracies decreased across generations for all three feature selection methods (GWAS, RF, and XGBoost). Over successive generations, the genomic landscape changes, and the features that were once predictive may no longer capture the genetic variation effectively. Notably, features selected by RF demonstrated better prediction accuracies in the second to fourth generations compared to those selected by GWAS. This suggests that RF is more effective at identifying features that retain predictive power over several generations.

XGBoost, which only outputs feature importance for the features used in model building, limited us to using approximately 3500 features. While these XGBoost-selected features initially showed better prediction accuracies compared to those selected by GWAS, this advantage diminished in later generations. The relatively low number of features selected by XGBoost likely contributed to this decline, as these features quickly broke down in subsequent generations.

The observed trends indicate that while RF is effective for short-term predictions, the stability of feature importance over generations is a critical factor in maintaining high prediction accuracies. The rapid degradation of XGBoost-selected features highlights the need for models that can identify a larger set of stable features to ensure long-term predictive power. These findings underscore the importance of continuously updating feature selection strategies to maintain robust prediction accuracies in breeding programs. Future research should focus on developing methods that combine the strengths of different feature selection techniques to enhance the robustness and longevity of genomic predictions across multiple generations.

4 | CONCLUSIONS

Using a large soybean dataset, we compared the prediction accuracies of features selected by GWAS against two machine learning approaches: RF and XGBoost. We found that GWAS performance could be compromised by high LD in the feature sets, while machine learning-based feature selection methods were less affected by LD. Therefore, LD pruning emerged as a crucial step in feature selection for GWAS. Among the two machine learning methods, RF proved to be more robust than XGBoost. XGBoost tends to produce a very small number of features, which perform well in the current generation but may perform poorly in future generations.

Using simulated data, we further explored the effect of feature selection over future breeding cycles. We found that, over successive generations, the genomic landscape changes,

and features that were once predictive may no longer capture genetic variation effectively. Consequently, it is crucial to continuously update and reassess feature selection strategies to maintain robust prediction accuracies in breeding programs. Future research should focus on developing adaptive methods that account for these changes and enhance the long-term utility of genomic predictions.

AUTHOR CONTRIBUTIONS

Hawladar A. Al-Mamun: Writing—original draft; writing—review and editing. **Monica F. Danilevicz:** Writing—review and editing. **Jacob I. Marsh:** Writing—review and editing. **Cedric Gondro:** Writing—review and editing. **David Edwards:** Writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Of these lines, 118 were previously published in PRJNA257011 (Fang et al., 2017) and 104 were previously published in PRJNA289660 (Valliyodan et al., 2016). The rest of the sequenced data are publicly available from the SRA project PRJNA639876 (Bayer et al., 2022).

ORCID

Hawladar A. Al-Mamun  <https://orcid.org/0000-0003-2453-0914>

David Edwards  <https://orcid.org/0000-0001-7599-6760>

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